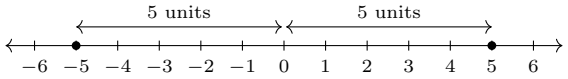
Lesson 1: Absolute Value of Real Numbers

Absolute value as a distance from zero, evaluating and ordering expressions that contain absolute values, the distance between two numbers, and the piecewise definition.

1. Absolute Value as a Distance from Zero

The **absolute value** of a real number a, written $|a|$, is the distance from a to 0 on a number line. A distance is never negative, so $|a| \geq 0$ for every real number. A number and its opposite are the same distance from 0, so they have the same absolute value: $|5| = 5$ and $|-5| = 5$.



Squaring a number removes its sign, and the principal square root is never negative, so the absolute value can also be found as $\sqrt{a^2}$. For example, $\sqrt{(-7)^2} = \sqrt{49} = 7 = |-7|$.

Absolute value: $|a|$ is the distance from a to 0. $\quad |a| \geq 0 \quad |-a| = |a| \quad |a| = \sqrt{a^2}$

The bars act only on what is inside them. In $-|6|$ the negative sign is outside the bars, so $-|6| = -6$; in $|-6|$ it is inside, so $|-6| = 6$.

Example 1

Evaluate.

(a) $|-14|$ (b) $|0|$ (c) $-|11|$ (d) $-|-2.5|$

(e) $\left|-\frac{7}{3}\right|$ (f) $|-\sqrt{5}|$ (g) $\sqrt{(-9)^2}$ (h) $-\sqrt{(-4)^2}$

2. Evaluating Expressions with Absolute Value

Absolute value bars are grouping symbols, like brackets: simplify the expression inside the bars first, then take its absolute value, then finish with the usual order of operations. The bars cannot be split over a sum or difference, so $|a + b|$ and $|a| + |b|$ are usually different. They can be split over a product or a quotient.

Order of operations: (1) simplify inside the bars; (2) take the absolute value; (3) finish: powers, then $\times$ and $\div$, then $+$ and $-$.

$|ab| = |a|\,|b|$ and $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, but in general $|a + b| \neq |a| + |b|$ and $|a - b| \neq |a| - |b|$.

On the TI-84 Plus CE, `abs(` is in the MATH menu under NUM. Type $|4 - 13|$ as `abs(4-13)`.

Example 2

Evaluate.

(a) $|4-13|$ (b) $|4|-|13|$ (c) $|-6+2|$ (d) $|-6|+|2|$

(e) $3|-5|-2|4|$ (f) $-2|1-7|+|-3|^2$ (g) $\big||2-9|-|-11|\big|$ (h) $|(-3)(4)+5|$

Example 3

Evaluate. In (d), give the exact value and the value to the nearest hundredth.

(a) $|\sqrt[3]{-64}|$ (b) $-|-\sqrt[3]{125}|$ (c) $|\sqrt{16}-\sqrt{81}|$ (d) $|2-\sqrt{10}|$

Evaluate each expression for $x=-3$.

(e) $|2x-7|$ (f) $|x|-|x-5|$ (g) $|x^2-10|$ (h) $-4|x+1|$

3. Comparing and Ordering

To compare or order expressions that contain absolute values, evaluate each one first (a decimal approximation is fine for ordering), then compare the values. To show that a statement is false, one **counterexample** (a set of values that makes it false) is enough.

Example 4

Arrange in order from least to greatest. $|-3.6|$, $-|4|$, $|2-7|$, $-|-1.5|$, $\left|\dfrac{-10}{3}\right|$, $|-\sqrt{12}|$

Example 5

(a) Let $a = -4$ and $b = 6$. Compare $|a + b|$ with $|a| + |b|$, and $|ab|$ with $|a|\,|b|$.

(b) True or false? If true, explain; if false, give a counterexample.

(i) $|3 - 10| = 3 - 10$

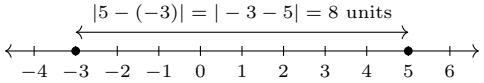
(ii) $|-2| \times |6| = |-12|$

(iii) $|x| = x$ for every real number x

(iv) $|a - b| = |b - a|$ for all real a and b

4. The Distance Between Two Numbers

On a number line, the distance between a and b is the absolute value of their difference. Subtracting in either order gives the same distance.



Distance between a and b: $|a - b| = |b - a|$. The numbers at a distance d from a are $a + d$ and $a - d$.

Absolute value is used whenever only the size of a change or a difference matters, not its direction: a change in temperature, the number of floors an elevator travels, or how far a measurement is from its target.

Example 6

Determine the distance between the numbers. Give exact answers.

(a) 9 and 2

(b) -6 and 4

(c) -11 and -3

(d) -2.75 and 1.5

(e) $\sqrt{2}$ and $-\sqrt{8}$

(f) Which numbers are 4 units from -1?

Example 7

(a) The overnight low in Airdrie was $-17\,^{\circ}\text{C}$ and the next afternoon's high was $4\,^{\circ}\text{C}$. By how many degrees did the temperature change?

(b) An elevator starts on floor 14. Its next four trips are $+7$, -15, $+4$ and -9 floors (up is positive). On which floor does it finish, and how many floors does it travel in all?

(c) A machine cuts bolts with a target length of 45.00 mm. A bolt of length L mm is accepted if $|L - 45.00| \leq 0.15$. Which of these bolts are accepted: 44.82 mm, 45.09 mm, 45.15 mm, 45.21 mm?

5. The Piecewise Definition

The distance idea gives an algebraic rule. A number that is positive or zero is its own absolute value. A negative number's absolute value is its opposite: $|-6| = -(-6) = 6$. Written in two pieces:

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

In the second piece x is negative, so $-x$ is **positive**. For every real number x, $\sqrt{x^2} = |x|$.

In Unit 2 we wrote $\sqrt{x^2} = x$ only after stating $x \geq 0$; without that restriction, $\sqrt{x^2} = |x|$. The same definition works for an expression inside the bars: $|x - 4|$ is the distance between x and 4, and it equals $x - 4$ when $x - 4 \geq 0$ and $-(x - 4)$ when $x - 4 < 0$. Lesson 2 does this for every expression $ax + b$.

Example 8

(a) Use the definition to evaluate $|-3.8|$ and $|2.4|$.

(b) If $x < 0$, simplify $|x| + 3x$.

(c) Simplify $\dfrac{|x|}{x}$ for $x > 0$ and for $x < 0$.

(d) Show that $\sqrt{x^2} = |x|$ when $x = -5$, and explain why $\sqrt{x^2} = x$ is false for $x = -5$.

(e) Write $|x - 4|$ in piecewise form. Use it to evaluate $|x - 4|$ for $x = 10$ and for $x = 1$.

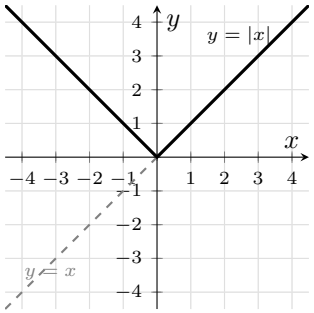
Lesson 2: Graphing Absolute Value Functions of Linear Functions

Graphing $y = |f(x)|$ by reflecting the part of a line below the x-axis, and describing its vertex, intercepts, domain, range and piecewise form.

1. The Graph of $y = |x|$

To graph the **absolute value function** $y = |x|$, compare its table with the table of $y = x$.

x	-3	-2	-1	0	1	2	3
$y = x$	-3	-2	-1	0	1	2	3
$y = \|x\|$	3	2	1	0	1	2	3



For $x \geq 0$ the y-values are the same. For $x < 0$ the y-values of $y = x$ are negative and those of $y = |x|$ are their opposites, so those points are **reflected in the x-axis**.

The graph of $y = |x|$ is a V shape with its **vertex** at $(0, 0)$. It is made of two half-lines, the two pieces of the definition: $y = x$ for $x \geq 0$ and $y = -x$ for $x < 0$.

Domain: $\{x \mid x \in \mathbb{R}\}$ Range: $\{y \mid y \geq 0, y \in \mathbb{R}\}$

Example 1

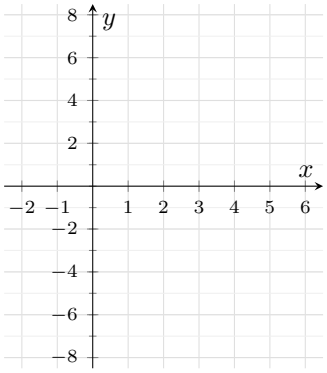
Let $f(x) = 2x - 4$.

(a) Complete the table.

x	-1	0	1	2	3	4	5
$y = f(x)$							
$y = \|f(x)\|$							

(b) Graph $y = f(x)$ and $y = |f(x)|$ on the grid.

(c) For which x-values are the two graphs the same? What happens to the other points?



2. Graphing $y = |f(x)|$ from the Graph of $y = f(x)$

Example 1 shows the rule for any function $y = f(x)$.

To graph $y = |f(x)|$ from $y = f(x)$:

- where the graph of $y = f(x)$ is on or above the x-axis ($f(x) \geq 0$), keep it;
- where it is below the x-axis ($f(x) < 0$), reflect it in the x-axis: $(x, y) \to (x, -y)$.

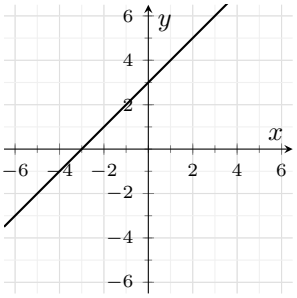
The graph of $y = |f(x)|$ is never below the x-axis. Points on the x-axis do not move: the x-intercepts of $y = f(x)$ are the **invariant points**.

A non-horizontal line $y = ax + b$ crosses the x-axis once, so the graph of $y = |ax + b|$ is a V shape whose **vertex** is that x-intercept. To check a sketch on the calculator, graph `Y1 = abs(2X-4)` (`abs(` is in the MATH menu under NUM).

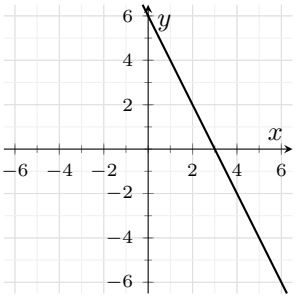
Example 2

The graph of $y = f(x)$ is shown. Sketch the graph of $y = |f(x)|$ on the same grid and state the invariant point.

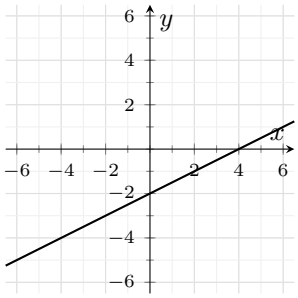
(a) $f(x) = x + 3$



(b) $f(x) = -2x + 6$



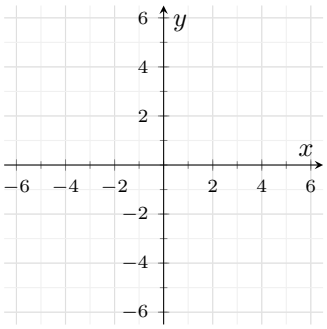
(c) $f(x) = \frac{1}{2}x - 2$



Example 3

(a) Sketch $y = |4 - 2x|$ without a table: find the vertex, the y-intercept and one more point.

(a) $y = |4 - 2x|$



(b) Let $g(x) = -2$ and $h(x) = 3$ (horizontal lines). Describe the graphs of $y = |g(x)|$ and $y = |h(x)|$.

3. Characteristics of $y = |ax + b|$

For $y = |ax + b|$, $a \neq 0$:

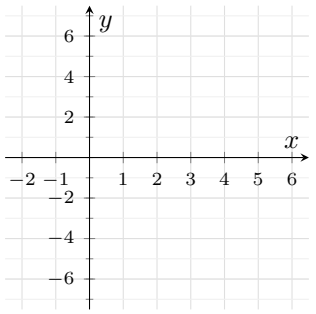
- vertex (the x-intercept and invariant point): solve $ax + b = 0$, giving $\left(-\frac{b}{a}, 0\right)$
- y-intercept: $(0, |b|)$
- domain $\{x \mid x \in \mathbb{R}\}$; range $\{y \mid y \geq 0, y \in \mathbb{R}\}$
- the two arms have slopes a and $-a$; the graph is symmetric about the vertical line through the vertex.

Compared with the line $y = ax + b$: the domain and the x-intercept stay the same, the range changes from all real numbers to $y \geq 0$, and the y-intercept b becomes $|b|$.

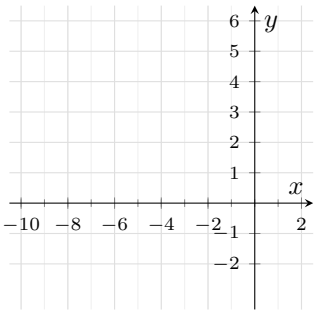
Example 4

(a) Let $f(x) = 3x - 6$. Complete the table, then sketch $y = f(x)$ and $y = |f(x)|$ on the grid.

	$y = f(x)$	$y = \|f(x)\|$
Domain		
Range		
x-intercept		
y-intercept		



(b) For $y = \left|-\frac{1}{2}x - 2\right|$, state the vertex, the y-intercept, the domain and the range, and the slopes of the arms. Sketch the graph.



4. Piecewise Form of $y = |ax + b|$

Apply the definition of absolute value to the whole expression $ax + b$:

$$|ax+b| = \begin{cases} ax+b, & \text{if } ax+b \geq 0 \\ -(ax+b), & \text{if } ax+b < 0 \end{cases}$$

Solve $ax + b \geq 0$ for x to see where each piece applies. **Dividing by a negative number reverses the inequality sign.**

The boundary is the x-coordinate of the vertex. At the vertex both pieces equal 0, so the equal sign can go with either piece; we put it with the first. On the graph, the piece $ax + b$ is the original line (above the axis) and the piece $-(ax + b)$ is the reflected arm.

Example 5

Write each function in piecewise form.

(a) $y = |x + 5|$

(b) $y = |2x - 7|$

(c) $y = |6 - 3x|$

(d) $y = \left|-\frac{1}{2}x + 1\right|$

Example 6

True or false? Explain.

(a) $|x-4| = x-4$ if $x \geq 4$

(b) $|3x+6| = 3x+6$ if $x < -2$

(c) $|5-x| = x-5$ if $x > 5$

(d) $|2x-3| = -2x+3$ if $x \leq \frac{3}{2}$

5. Reasoning with $y = |f(x)|$

- A point (x, y) on $y = f(x)$ becomes the point $(x, |y|)$ on $y = |f(x)|$.
- If (x, k) with $k > 0$ is on $y = |f(x)|$, then $f(x) = k$ or $f(x) = -k$: the point on $y = f(x)$ is (x, k) or $(x, -k)$.
- For a linear function, the same V comes from two lines: $y = f(x)$ and its reflection $y = -f(x)$.
- $|f(x)|$ is never negative, so an equation such as $|f(x)| = -4$ has no solution and $|f(x)| < 0$ is never true.

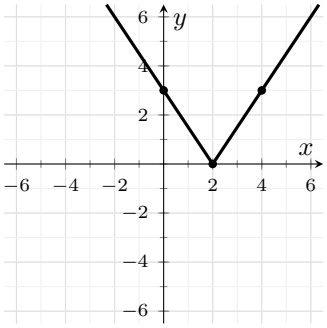
Example 7

(a) The point $(-4, -7)$ is on the graph of $y = f(x)$. State the matching point on $y = |f(x)|$.

(b) The point $(2, 5)$ is on the graph of $y = |g(x)|$. State the possible points on $y = g(x)$.

(c) The graph of $y = |f(x)|$, where f is linear, is shown. Sketch the two possible graphs of $y = f(x)$ and write their equations. Then write $y = |f(x)|$ in the form $y = |ax + b|$.

$y = |f(x)|$



(d) Explain why the equation $|f(x)| = -4$ has no solution.

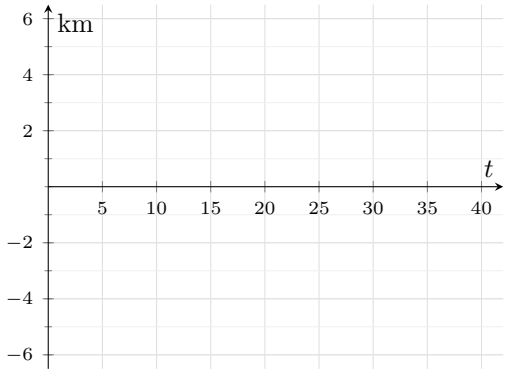
Example 8

A cyclist rides at a constant 0.25 km/min along a straight trail toward a water station 5 km ahead, rides past it, and keeps going. Her position relative to the station after t minutes is $p = 5 - 0.25t$ kilometres (positive before the station, negative after it), and her distance from the station is $d = |5 - 0.25t|$, for $0 \le t \le 40$.

(a) Complete the table.

t (min)	0	10	20	30	40
p (km)					
d (km)					

(b) Graph p and d on the grid.



(c) When does she reach the station?

(d) Use the graph to find when she is 2 km from the station.

(e) State the domain and the range of d in this context.

Lesson 3: Graphing Absolute Value Functions of Quadratic Functions

Graphing $y = |f(x)|$ for a quadratic function by reflecting the part of the parabola below the x-axis, and describing its invariant points, domain, range and piecewise form.

1. From $y = f(x)$ to $y = |f(x)|$ for a Parabola

The rule from Lesson 2 works for every function: keep the part of the graph of $y = f(x)$ that is on or above the x-axis, and reflect the part below the x-axis in the x-axis, $(x, y) \to (x, -y)$. Each y-value of $y = |f(x)|$ is the absolute value of the y-value of $y = f(x)$ at the same x.

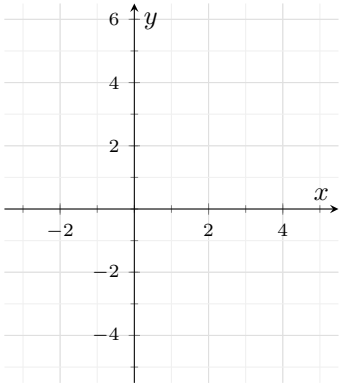
For a parabola that crosses the x-axis twice, the part below the axis is a single arc: the arc **between the zeros** when the parabola opens up, and the two arms **outside the zeros** when it opens down. After the reflection the graph has a sharp corner at each x-intercept, and the vertex of that arc is reflected too.

Example 1

Consider $f(x) = x^2 - 2x - 3$.

(a) Complete the table of values.

x	$y = f(x)$	$y = \lvert f(x) \rvert$
-2		
-1		
0		
1		
2		
3		
4		



(b) Plot both graphs on the grid.
(c) Which points are the same on both graphs? Which part of $y = f(x)$ was reflected?

To sketch $y = |f(x)|$ for a quadratic function f:
(1) sketch $y = f(x)$ (vertex, x-intercepts, y-intercept); (2) keep every part on or above the x-axis;
(3) reflect every part below the x-axis in the x-axis: the y-values change sign, the x-values stay;
(4) the x-intercepts are the **invariant points**; a vertex (p, q) below the axis moves to $(p, -q)$.

2. Sketching $y = |f(x)|$ from a Graph or an Equation

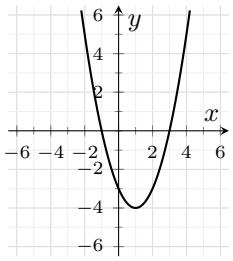
Where the parabola sits decides how much of it changes:

- it crosses the x-axis: only the part below the axis is reflected, and the graph has two invariant points;
- it is entirely above the x-axis: nothing changes, $|f(x)| = f(x)$;
- it is entirely below the x-axis: the whole parabola is reflected, $|f(x)| = -f(x)$, and there are no invariant points.

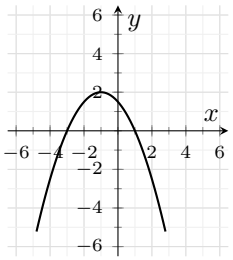
Example 2

The graph of $y = f(x)$ is shown. Sketch the graph of $y = |f(x)|$ on the same grid. State the invariant points, if any.

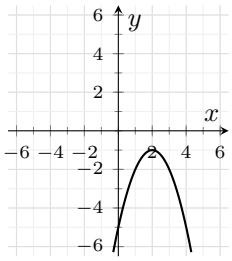
(a)



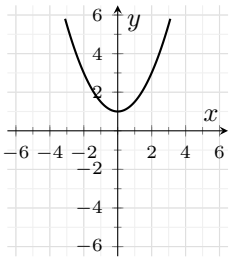
(b)



(c)



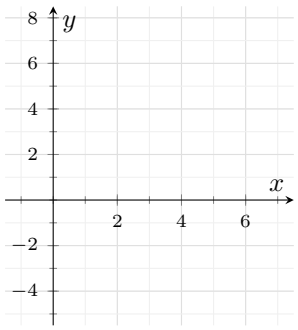
(d)

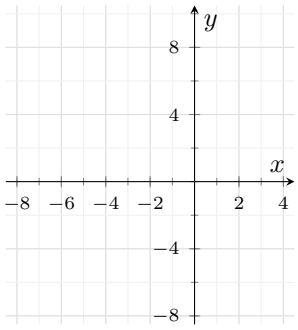


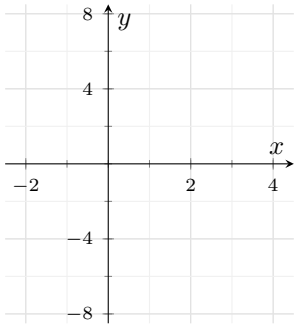
Example 3

Sketch the graph of each function. Show the graph of $y = f(x)$ lightly first, and label the invariant points and the vertex of $y = |f(x)|$.

(a) $y = |x^2 - 6x + 5|$ (b) $y = \left|-\frac{1}{2}(x+2)^2 + 8\right|$ (c) $y = |-(x-1)^2 - 2|$







3. Invariant Points, Domain and Range

Reflecting in the x-axis does not change any x-value, so the domain of $y = |f(x)|$ is the same as the domain of $y = f(x)$: all real numbers for a quadratic function. The range changes, because no y-value of $y = |f(x)|$ is negative.

For a quadratic function $f(x) = a(x-p)^2 + q$**:** domain of $y = |f(x)|$: $\{x \mid x \in \mathbb{R}\}$ y-intercept: $|f(0)|$

The parabola ...	Invariant points	Range of $y = \|f(x)\|$
crosses or touches the x-axis	the x-intercepts	$\{y \mid y \geq 0, y \in \mathbb{R}\}$
is entirely above the x-axis ($a > 0$, $q > 0$)	none	$\{y \mid y \geq q, y \in \mathbb{R}\}$ (unchanged)
is entirely below the x-axis ($a < 0$, $q < 0$)	none	$\{y \mid y \geq -q, y \in \mathbb{R}\}$

When an arc between the zeros is reflected, its vertex $(p, -q)$ is the highest point of that arc but not of the whole graph, since the outer arms keep rising. The graph has no maximum value.

Example 4

Complete the table for each function without graphing it.

Function	Invariant points	y-intercept	Vertex of the graph	Range
(a) $y = \|x^2 + 2x - 8\|$				
(b) $y = \|2(x-3)^2 + 1\|$				
(c) $y = \|-3(x+1)^2 - 6\|$				
(d) $y = \|-(x-2)^2 + 9\|$				

4. Piecewise Form of $y = |f(x)|$

Without absolute value bars, $|f(x)| = f(x)$ where $f(x) \geq 0$ and $|f(x)| = -f(x)$ where $f(x) < 0$. For a parabola with zeros $r < s$, the sign of $f(x)$ changes only at the zeros, so find the zeros first and use the direction of opening (or a test value) to see where $f(x) < 0$.

If f opens **up** with zeros $r < s$: $\quad |f(x)| = \begin{cases} f(x), & \text{if } x \leq r \text{ or } x \geq s \\ -f(x), & \text{if } r < x < s \end{cases}$

If f opens **down**, the pieces swap: $|f(x)| = f(x)$ for $r \leq x \leq s$ and $-f(x)$ for $x < r$ or $x > s$.
If $f(x) \geq 0$ for every x, one piece is enough: $|f(x)| = f(x)$. If $f(x) < 0$ for every x, $|f(x)| = -f(x)$.

The piecewise form can also be written in three pieces (left arm, middle arc, right arm). At a zero both pieces give 0, so it does not matter which piece includes it.

Example 5

Write each function in piecewise form, without absolute value symbols.

(a) $y = |x^2 - x - 12|$

(b) $y = |16 - 4x^2|$

(c) $y = |x^2 + 4x + 7|$

(d) Use (a) to evaluate y at $x = 1$ and at $x = 5$. Check with the bars.

Example 6

True or false? Explain.

(a) $|x^2 - 1| = 1 - x^2$ if $-1 \leq x \leq 1$

(b) $|x^2 - 25| = x^2 - 25$ if $x > -5$

(c) $|(x - 6)^2| = (x - 6)^2$ for every real x

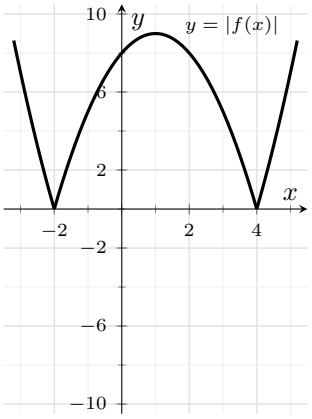
(d) $|-x^2 + 4| = |x^2 - 4|$ for every real x

5. Working Backwards and Solving Problems

The graph of $y = |f(x)|$ does not show which parts of $y = f(x)$ were below the axis. If f is quadratic, there are two possibilities: $f(x)$ or its reflection $-f(x)$, because both have the same absolute value. A point (x, k) on $y = |f(x)|$ with $k > 0$ comes from (x, k) or $(x, -k)$ on $y = f(x)$.

Example 7

(a) The graph of $y = |f(x)|$ is shown, where f is a quadratic function. Sketch the two possible graphs of $y = f(x)$ and write the equation of each in factored form.

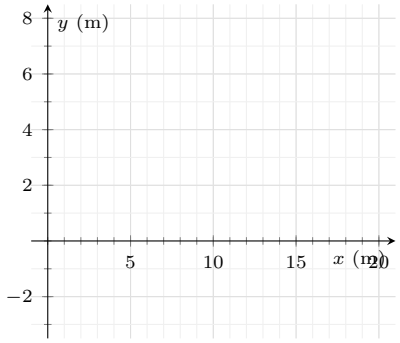


(b) $g(x) = -2(x-1)^2 + k$. Determine k if the range of $y = |g(x)|$ is $\{y \mid y \geq 4, y \in \mathbb{R}\}$.

(c) For which values of k is the range of $y = |g(x)|$ equal to $\{y \mid y \geq 0, y \in \mathbb{R}\}$?

Example 8

Part of a roller coaster track dips into a tunnel below ground level. Its height above ground level is modelled by $h(x) = 0.1x^2 - 2x + 7.5$, where h is in metres and x is the horizontal distance in metres, $0 \leq x \leq 20$. The distance between the track and ground level is $d(x) = |h(x)|$.
(a) Where is the track at ground level?



(b) Sketch $y = d(x)$ on the grid. (c) How far below ground does the track go, and where?

(d) State the range of d for $0 \leq x \leq 20$.

Lesson 4: Solving Absolute Value Equations (Linear)

Solving equations such as $|2x - 5| = 9$ and $|x - 4| = 3x$ graphically and algebraically by cases, and rejecting extraneous solutions.

1. What an Absolute Value Equation Means

An **absolute value equation** has the variable inside absolute value bars, for example $|x - 3| = 5$ or $|2x + 3| = x + 6$. Since $|x - 3|$ is the distance between x and 3, the equation $|x - 3| = 5$ asks for the numbers 5 units from 3: $x = -2$ or $x = 8$.

An absolute value is never negative, so the value on the other side decides how many solutions there are.

For $\lvert A\rvert = c$, where c is a constant:
$c > 0$: two solutions, from $A = c$ or $A = -c$; $c = 0$: one solution, from $A = 0$; $c < 0$: **no solution**.

Example 1

Solve.

(a) $|x| = 7$ (b) $|x - 3| = 5$ (c) $|x + 4| = 0$ (d) $|x - 1| = -2$

2. Solving Graphically

A solution of an equation is an x-value that makes the two sides equal. On a graph these are the x-coordinates where the graphs of the two sides meet. A graphing calculator gives two ways to find them.

Intersection method: graph $y =$ left side and $y =$ right side; the solutions are the x-coordinates of the points of intersection (use INTERSECT).

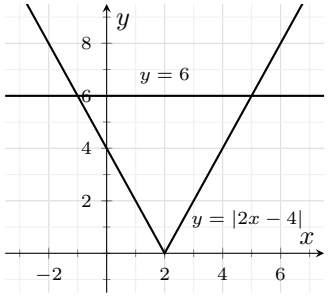
Zero method: move every term to one side, so the equation reads $(\dots) = 0$; graph $y = (\dots)$; the solutions are its zeros, the x-intercepts (use ZERO).

On the TI-84 Plus CE, type $|2x - 4|$ as `abs(2X-4)` (MATH, NUM); INTERSECT and ZERO are in the CALC menu (2ND TRACE).

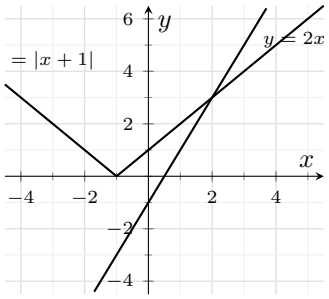
Example 2

Use the graphs to solve each equation. Then check each solution in the equation.

(a) $|2x - 4| = 6$



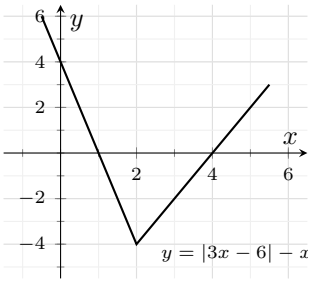
(b) $|x + 1| = 2x - 1$



Example 3

(a) Describe how to solve $|3x - 6| = x + 2$ with the zero method.

(b) The graph from (a) is shown. State the solution and check it.



3. Solving $|A| = c$ Algebraically

When the other side is a constant, first **isolate the absolute value** (undo the other operations on the outside of the bars). Then use the key idea from Section 1: $A = c$ or $A = -c$.

Example 4

Solve.

(a) $|2x - 5| = 9$

(b) $3|x + 2| - 4 = 11$

(c) $|4 - 3x| + 8 = 2$

(d) $-2|x - 6| + 1 = 1$

(e) $\frac{1}{2}|x + 3| = 4$

(f) A student solved $|x + 2| + 4 = 1$ this way: $x + 2 + 4 = 1$ or $x + 2 + 4 = -1$, so $x = -5$ or $x = -7$. Explain the error and give the correct solution.

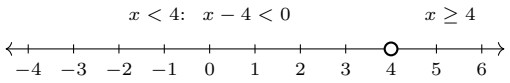
4. Solving $|A| = B$ by Cases

When the other side contains the variable, the number of solutions is not known in advance. Use the piecewise form of the absolute value (Lesson 2): on each side of the value that makes the inside zero, the bars can be replaced by an ordinary expression.

Solving by cases:
(1) find the value of x that makes the expression inside the bars equal to 0;
(2) split the number line there into two subdomains;
(3) in each subdomain, replace $|A|$ by A (where $A \geq 0$) or by $-A$ (where $A < 0$);
(4) solve the equation in each subdomain;
(5) keep a solution only if it is in its own subdomain; combine the solutions that are kept.

Example 5

Solve $|x - 4| = 3x$ by cases.



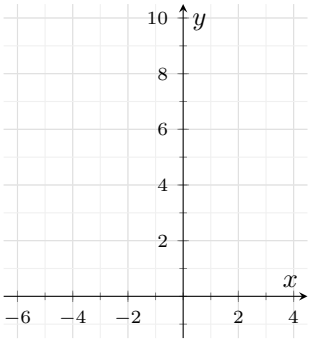
Case 1: $x \geq 4$, so $|x - 4| = x - 4$.

Case 2: $x < 4$, so $|x - 4| = -(x - 4)$.

Solution: _____ Check:

Example 6

(a) Solve $|2x + 3| = x + 6$ by cases.



(b) Sketch $y = |2x + 3|$ and $y = x + 6$ on the grid to confirm the solution.

5. Solving Both Equations, Then Verifying

A shorter method ignores the subdomains: write $A = B$ and $A = -B$, solve both, then **check every answer in the original equation**. An answer that fails the check is an **extraneous solution**: it solves one of the two linear equations but not the absolute value equation, usually because it makes the right side negative. Reject it.

To solve $|A| = B$: (1) isolate the absolute value; (2) solve $A = B$ and $A = -B$; (3) substitute each answer into the original equation; reject any that fail (the right side B must be ≥ 0).
There can be two solutions, one solution or no solution.

Example 7

Solve and verify.

(a) $|3x + 1| = x - 5$ (b) $2|x - 3| = x + 3$ (c) $|x + 5| - 2x = 1$

6. Problems

Absolute value equations describe numbers at a given distance from a target value: $|x - a| = d$ means "x is d units from a", so $x = a + d$ or $x = a - d$. Working backwards, two numbers m and n are the solutions of $|x - a| = d$ where $a = \dfrac{m + n}{2}$ is the midpoint and $d = \dfrac{|m - n|}{2}$ is half the distance between them.

Example 8

(a) A cereal company fills boxes with a target mass of 750 g. A box is rejected if its mass differs from the target by more than 12 g. Write and solve an absolute value equation for the least and greatest masses that are accepted.

(b) Write an absolute value equation, in the form $|x-a| = d$, whose solutions are -3 and 11.

(c) The distance between a number and 2 is twice the number. Write an equation and determine the number.

Lesson 5: Solving Absolute Value Equations (Quadratic)

Solving equations with the absolute value of a quadratic expression, or with a variable on both sides, by cases, by factoring or the quadratic formula, and by graphing, and rejecting extraneous solutions.

1. The Absolute Value of a Quadratic Equal to a Constant

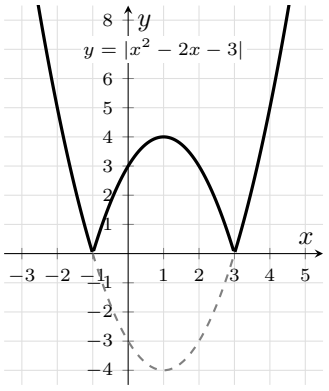
To solve $|A| = k$ where $k > 0$, use the two cases from Lesson 4: the expression inside is either k or $-k$. When A is a quadratic expression, each case is a quadratic equation, so there can be up to four solutions.

To solve $|f(x)| = k$**:** (1) isolate the absolute value; (2) if $k < 0$ there is no solution, and if $k = 0$ solve $f(x) = 0$; (3) if $k > 0$ solve **both** $f(x) = k$ and $f(x) = -k$ (each in standard form: factor, or use the quadratic formula); (4) a case with no real roots gives nothing; collect every real root.

Because the right side k is a positive constant, every root of the two cases is a solution (no root is extraneous). Checking one or two of them in the original equation is still a good habit.

Example 1

The graph of $y = |x^2 - 2x - 3|$ is shown. The part of the parabola $y = x^2 - 2x - 3$ between its zeros has been reflected.
(a) Draw the line $y = 5$ on the grid and use the intersection points to estimate the solutions of $|x^2 - 2x - 3| = 5$.



(b) Solve the equation algebraically.

Example 2

Solve algebraically.

(a) $|x^2 - 10| = 6$

(b) $|x^2 + 2x - 4| = 4$

(c) $2|x^2 - 5x| - 3 = 9$

(d) $|x^2 + 4| = 2$

2. Using the Quadratic Formula; How Many Solutions?

When a case does not factor, use the quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Give exact answers in simplest radical form, then round if asked.

The graph tells how many solutions to expect. If the parabola $y = f(x)$ opens up and its vertex (p, q) is below the x-axis, the graph of $y = |f(x)|$ is a W shape whose middle hump has its top at $(p, |q|)$.

Number of solutions of $|f(x)| = k$ (parabola crossing the x-axis, reflected hump of height $|q|$):

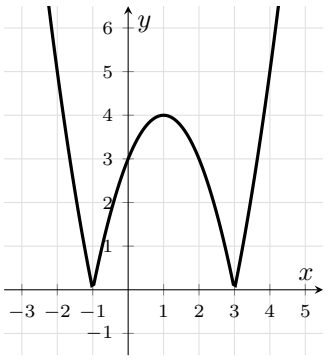
k	$k < 0$	$k = 0$	$0 < k < \lvert q\rvert$	$k = \lvert q\rvert$	$k > \lvert q\rvert$
solutions	0	2 (the zeros)	4	3	2

If the parabola does not cross the x-axis, the reflection changes nothing (or turns the whole graph over): there are 0, 1 or 2 solutions.

Example 3

Use the graph of $y = |x^2 - 2x - 3|$ from Example 1.

(a) Solve $|x^2 - 2x - 3| = 2$. Give exact answers and answers to the nearest hundredth.



(b) Solve $|x^2 - 2x - 3| = 4$ exactly. Why are there only three solutions?

Example 4

Consider $|x^2 - 6x + 5| = k$.

(a) In vertex form, $x^2 - 6x + 5 = (x - 3)^2 - 4$. Without solving, state the number of solutions when $k = -2$, $k = 0$, $k = 2$, $k = 4$ and $k = 7$.

(b) Solve the equation for $k = 7$ and confirm the number of solutions.

3. A Variable on the Other Side: Extraneous Solutions

In $|A| = B$ where B contains the variable (for example $|x^2 - 4| = 3x$), the right side could be negative for some x, and an absolute value is never negative. Solving the two cases can then produce **extraneous solutions**: values that solve a case but not the original equation.

To solve $|A| = B$ when B contains x: solve $A = B$ and $A = -B$, then **check every root in the original equation** (equivalently, keep only the roots that make $B \geq 0$). Reject the others as extraneous.

The method from Lesson 4 also works: split the number line where the expression inside the absolute value is 0, write $|A|$ as A or $-A$ on each part, solve, and keep only the roots that lie in their part. Either way, the check is required.

Example 5

Solve and check.

(a) $|x^2 - 4| = 3x$

(b) $|2x + 1| = x^2 - 2$

4. Equations with Two Absolute Values

Two numbers have the same absolute value when they are equal or opposites.

$|A| = |B| \iff A = B$ or $A = -B$. Both sides are never negative, so no root is extraneous.

For a sum or difference such as $|A| + |B| = k$, use **subdomains**: mark where each expression inside is 0 on a number line, write each absolute value without its bars on each interval (A if $A \geq 0$, $-A$ if $A < 0$), solve on each interval, and keep only roots that lie in that interval.

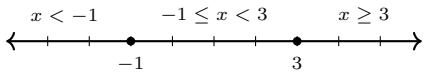
Example 6

Solve.
(a) $|x+5| = |2x-1|$

(b) $|x^2-3| = |2x|$

Example 7

Solve $|x-3| + |x+1| = 6$ using subdomains.



5. Solving Graphically and Solving Problems

A graphing calculator gives a quick check, or decimal answers when the algebra is messy.

Intersection method: graph Y_1 = the left side (`abs(` is in the MATH menu under NUM) and Y_2 = the right side; use **intersect** at each point where the graphs meet.
Zero method: graph Y_1 = (left side) − (right side) and use **zero** at each x-intercept.

Use a window that shows every intersection (the W shape and the whole line). In a word problem, an equation such as $|h-10| = 3$ means "h is exactly 3 away from 10", so $h = 13$ or $h = 7$.

Example 8

A drone rises and then lands. Its height is $h = -t^2 + 8t$ metres after t seconds, $0 \le t \le 8$. A camera on the ground records it only while it is within 3 m of a height of 10 m.

(a) Explain why the times when the drone is exactly 3 m from a height of 10 m are the solutions of $|-t^2+8t-10| = 3$. Solve the equation exactly, then to the nearest hundredth of a second.

(b) Check with a graphing calculator, then state the time intervals during which the camera records the drone.

Lesson 6: Reciprocal Functions of Linear Functions

Graphing $y = \frac{1}{f(x)}$ from a line $y = f(x)$ using its asymptotes and invariant points, and stating its domain and range.

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1. Reciprocals and the Graph of $y = \frac{1}{x}$

The **reciprocal** of a non-zero number a is $\frac{1}{a}$. A number and its reciprocal have the same sign; the reciprocal of a large number is close to 0, the reciprocal of a number close to 0 is large, 1 and -1 are their own reciprocals, and 0 has no reciprocal.

The **reciprocal function** of $y = f(x)$ is $y = \frac{1}{f(x)}$. For example, the reciprocal of $f(x) = x$ is $y = \frac{1}{x}$, and the reciprocal of $f(x) = 2x + 5$ is $y = \frac{1}{2x+5}$.

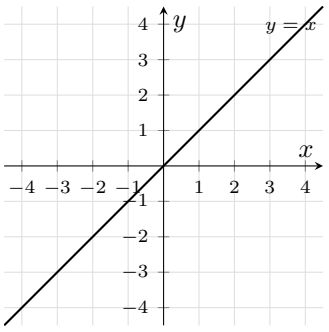
Example 1

(a) Complete the table for $y = \frac{1}{x}$.

x	-4	-2	-1	$-\frac{1}{2}$	$-\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
$\frac{1}{x}$											

(b) The line $y = x$ is drawn. Sketch $y = \frac{1}{x}$ on the same grid.

(c) State the points where the two graphs meet, the asymptotes of $y = \frac{1}{x}$, and its domain and range.



2. Comparing $y = f(x)$ and $y = \dfrac{1}{f(x)}$

Each y-value of $y = \dfrac{1}{f(x)}$ is the reciprocal of the y-value of $y = f(x)$ at the same x, so a point (x, y) on $y = f(x)$ becomes $\left(x, \dfrac{1}{y}\right)$ (when $y \neq 0$).

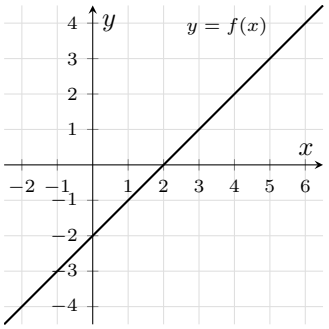
Example 2

Let $f(x) = x - 2$.

(a) Complete the table.

x	-2	0	1	1.5	2	2.5	3	4	6
$f(x)$									
$\frac{1}{f(x)}$									

(b) Sketch $y = \dfrac{1}{f(x)}$ on the grid with $y = f(x)$.



(c) Complete the statements.

y-intercept of $y = f(x)$: _____ of $y = \frac{1}{f(x)}$: _____

x-intercept of $y = f(x)$: _____ vertical asymptote: _____

invariant points: _____ horizontal asymptote: _____

Example 3

Use $y = \dfrac{1}{x - 2}$ from Example 2 to describe the graph near its asymptotes.

x	1.9	1.99	1.999	2.001	2.01	2.1	-98	102
$\frac{1}{x-2}$								

These examples show what happens for any function.

Properties of $y = \dfrac{1}{f(x)}$:

- Where $f(x) = 0$, the reciprocal is undefined: a **vertical asymptote** (for a line, at its x-intercept).
- Where $f(x) = 1$ or $f(x) = -1$, the reciprocal is the same number: the **invariant points** lie on the lines $y = 1$ and $y = -1$.
- $\dfrac{1}{f(x)}$ has the same sign as $f(x)$: both graphs are above the x-axis, or both are below, at each x.
- Where $f(x)$ increases, $\dfrac{1}{f(x)}$ decreases, and the other way around.
- Where $|f(x)|$ is large, $\dfrac{1}{f(x)}$ is near 0: the **horizontal asymptote** is $y = 0$. A reciprocal is never 0, so the graph has no x-intercept.
- Where $f(x)$ is near 0, $\left|\dfrac{1}{f(x)}\right|$ is large: the graph runs along the vertical asymptote.
- The y-intercept of $y = \dfrac{1}{f(x)}$ is the reciprocal of the y-intercept of $y = f(x)$; a point where $f(x) = 2$ becomes a point where $y = \frac{1}{2}$, and so on.

3. Sketching $y = \dfrac{1}{f(x)}$ from the Graph of a Line

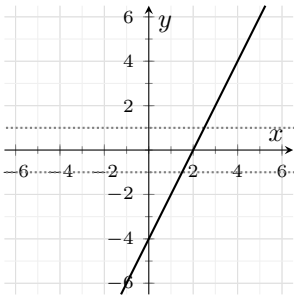
To sketch $y = \dfrac{1}{f(x)}$ from the line $y = f(x)$: (1) draw the vertical asymptote (dashed) through the x-intercept, and the horizontal asymptote $y = 0$; (2) mark the invariant points where the line meets $y = 1$ and $y = -1$; (3) mark a few more points, such as the reciprocal of the y-intercept, or $y = \frac{1}{2}$ where $f(x) = 2$; (4) draw two smooth branches, one on each side of the vertical asymptote, on the same side of the x-axis as the line, approaching both asymptotes.

The graph of the reciprocal of a non-horizontal line always has two branches in diagonally opposite regions, like $y = \dfrac{1}{x}$.

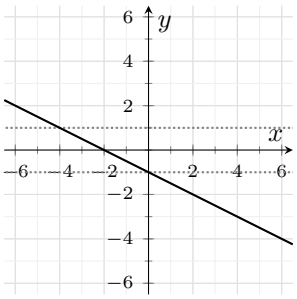
Example 4

The graph of $y = f(x)$ and the guide lines $y = \pm 1$ are shown. Sketch $y = \dfrac{1}{f(x)}$. State the equation of the vertical asymptote, the invariant points and the y-intercept of the reciprocal graph.

(a) $f(x) = 2x - 4$



(b) $f(x) = -\frac{1}{2}x - 1$



4. Characteristics from the Equation

For $y = \dfrac{1}{ax + b}$, $a \neq 0$, find each feature without graphing:

- vertical asymptote: solve $ax + b = 0$, $x = -\dfrac{b}{a}$
- horizontal asymptote: $y = 0$
- invariant points: solve $ax + b = 1$ and $ax + b = -1$
- y-intercept: $\dfrac{1}{b}$ (if $b \neq 0$)
- domain $\left\{x \mid x \neq -\dfrac{b}{a}, x \in \mathbb{R}\right\}$; range $\{y \mid y \neq 0, y \in \mathbb{R}\}$

To check on a graphing calculator, enter `Y1 = 1/(3X-6)` with brackets around the whole denominator.

Example 5

Complete the table.

	(a) $y = \dfrac{1}{x+4}$	(b) $y = \dfrac{1}{3x-6}$	(c) $y = \dfrac{1}{5-2x}$
Vertical asymptote			
y-intercept			
Invariant points			
Domain			
Range			
$y > 0$ when			

Example 6

(a) The graph of $y = f(x)$ passes through $(5, -4)$. Which point with $x = 5$ is on the graph of $y = \dfrac{1}{f(x)}$?

(b) The graph of $y = \dfrac{1}{f(x)}$ passes through $(-3, 0.2)$. Which point with $x = -3$ is on the graph of $y = f(x)$?

(c) For $f(x) = \frac{1}{2}x + 1$, determine the point on $y = \dfrac{1}{f(x)}$ where $y = 4$.

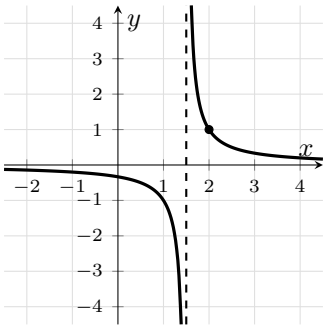
(d) Can the graph of $y = \dfrac{1}{f(x)}$ ever have $y = 0$? Explain.

5. Working Backwards and Matching

The features go both ways: a vertical asymptote $x = a$ of $y = \dfrac{1}{f(x)}$ is a zero of f, an invariant point is on both graphs, and a point $\left(x, \dfrac{1}{y}\right)$ on the reciprocal gives (x, y) on the line. Two points determine the line $y = f(x)$.

Example 7

The graph of $y = \dfrac{1}{f(x)}$, where $f(x)$ is a linear function, is shown. It has the vertical asymptote $x = 1.5$ and passes through the invariant point $(2, 1)$.



(a) Sketch the graph of $y = f(x)$ on the grid.

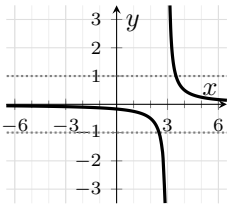
(b) Determine the equation of $f(x)$ in the form $f(x) = ax + b$, and the y-intercept of $y = \dfrac{1}{f(x)}$.

Example 8

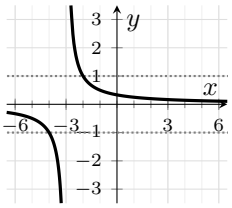
Match each equation with its graph. Explain how you decided.

(i) $y = \dfrac{1}{x+3}$ (ii) $y = \dfrac{1}{2x-6}$ (iii) $y = \dfrac{1}{x-3}$ (iv) $y = \dfrac{1}{3-x}$

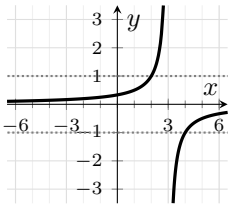
A



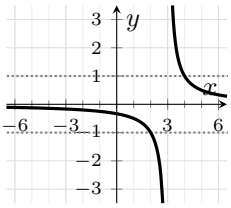
B



C



D



Lesson 7: Reciprocal Functions of Quadratic Functions

Graphing $y = \dfrac{1}{f(x)}$ when f is quadratic: zero, one or two vertical asymptotes, the turning point at the reciprocal of the vertex, invariant points, domain and range.

1. The Reciprocal Rules, Applied to a Parabola

The rules from Lesson 6 work for any function. A parabola just gives the reciprocal graph more features.

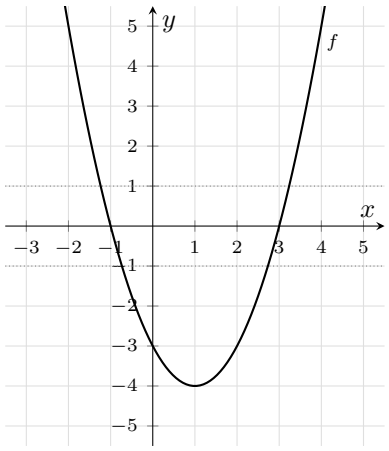
From $y = f(x)$ to $y = \dfrac{1}{f(x)}$:

- each point (x, y) with $y \neq 0$ becomes $\left(x, \frac{1}{y}\right)$; the y-intercept $(0, c)$ becomes $\left(0, \frac{1}{c}\right)$
- each **zero** of f gives a **vertical asymptote** $x = a$ of the reciprocal graph
- $\frac{1}{f(x)}$ is never 0; as $f(x)$ grows large (positive or negative), $\frac{1}{f(x)}$ approaches 0: the **horizontal asymptote** is $y = 0$
- where $f(x) = 1$ or $f(x) = -1$, the reciprocal is the same number: these are the **invariant points** (on the lines $y = \pm 1$)
- $\frac{1}{f(x)}$ has the same sign as $f(x)$: the reciprocal graph is above the x-axis exactly where f is
- where f increases, $\frac{1}{f}$ decreases, and where f decreases, $\frac{1}{f}$ increases (on each interval between asymptotes)

To check a sketch on the calculator, enter the quadratic as `Y1` and `Y2 = 1/Y1`, and use TABLE to read values.

Example 1

Let $f(x) = x^2 - 2x - 3$. Its graph is shown, with the lines $y = \pm 1$ dotted.



(a) Complete the table.

x	-2	-1	0	1	2	3	4
$f(x)$							
$\frac{1}{f(x)}$							

(b) Plot the points and sketch $y = \dfrac{1}{f(x)}$ on the grid.

(c) State the equations of the asymptotes.

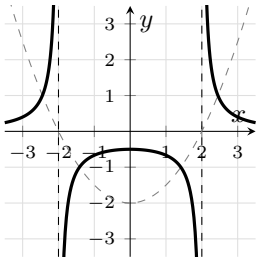
(d) The vertex of f is $(1, -4)$. Which point on the reciprocal graph matches it? Explain why it is the highest point of the middle branch.

(e) Use the dotted lines to estimate the four invariant points. Then find them to the nearest hundredth by solving $f(x) = 1$ and $f(x) = -1$.

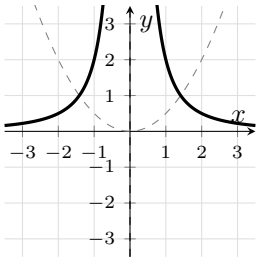
2. How Many Vertical Asymptotes?

The vertical asymptotes of $y = \dfrac{1}{f(x)}$ are at the zeros of f. A quadratic function has two, one or no real zeros, so its reciprocal has **two, one or no vertical asymptotes**. Find the zeros by factoring or the quadratic formula, or count them with the discriminant $b^2 - 4ac$ (Unit 5).

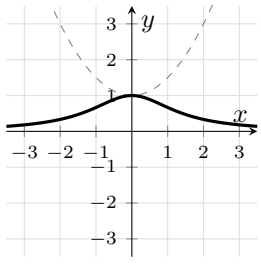
$f(x) = \frac{1}{2}x^2 - 2$



$f(x) = \frac{1}{2}x^2$



$f(x) = \frac{1}{2}x^2 + 1$



Zeros of f $\rightarrow$ vertical asymptotes of $y = \dfrac{1}{f(x)}$ (f dashed, its reciprocal solid, above)

- **Two zeros** ($b^2 - 4ac > 0$): two vertical asymptotes; three branches, and the middle one has a turning point.
- **One zero** ($b^2 - 4ac = 0$, the vertex is on the x-axis): one vertical asymptote; two branches on the same side of the x-axis.
- **No zeros** ($b^2 - 4ac < 0$): no vertical asymptote; one unbroken curve, entirely above or entirely below the x-axis.

Example 2

State the number of vertical asymptotes of the graph of $y = \dfrac{1}{f(x)}$ and their equations.

(a) $f(x) = x^2 + 3x - 10$ (b) $f(x) = x^2 - 10x + 25$ (c) $f(x) = 3x^2 + 2$

(d) $f(x) = 2x^2 + 5x - 3$ (e) $f(x) = x^2 - 2x + 4$ (f) $f(x) = x^2 - 7$

3. The Turning Point, the Domain and the Range

On an interval where $f(x)$ keeps the same sign, $\dfrac{1}{f(x)}$ is largest where $f(x)$ is smallest, and smallest where $f(x)$ is largest. So the vertex of the parabola becomes the turning point of the reciprocal graph, with its type switched.

Turning point. If the vertex of $f(x) = a(x-p)^2 + q$ is (p, q) with $q \neq 0$, the graph of $y = \dfrac{1}{f(x)}$ has a turning point at $\left(p, \frac{1}{q}\right)$:
• $a > 0$ (minimum of f) $\rightarrow$ a **maximum point** of the reciprocal graph; $a < 0$ (maximum of f) $\rightarrow$ a **minimum point**.
• If $q = 0$ the vertex is a zero: there is a vertical asymptote $x = p$ and no turning point. If $q = 1$ or $q = -1$ the vertex is an invariant point.

Domain: every real number except the zeros of f. **Range:** take the reciprocal of each part of the range of f (never $y = 0$).

Range of $y = \dfrac{1}{f(x)}$	$q > 0$	$q < 0$	$q = 0$
$a > 0$ (opens up)	$\left\{y \mid 0 < y \leq \frac{1}{q}, y \in \mathbb{R}\right\}$	$\left\{y \mid y \leq \frac{1}{q} \text{ or } y > 0, y \in \mathbb{R}\right\}$	$\{y \mid y > 0, y \in \mathbb{R}\}$
$a < 0$ (opens down)	$\left\{y \mid y < 0 \text{ or } y \geq \frac{1}{q}, y \in \mathbb{R}\right\}$	$\left\{y \mid \frac{1}{q} \leq y < 0, y \in \mathbb{R}\right\}$	$\{y \mid y < 0, y \in \mathbb{R}\}$

You do not need to memorize the table. For example, if f has range $y \geq -2$: the values $-2 \leq f(x) < 0$ have reciprocals $\dfrac{1}{f(x)} \leq -\dfrac{1}{2}$, and the values $f(x) > 0$ have reciprocals $\dfrac{1}{f(x)} > 0$.

Example 3

For each function, state the turning point of $y = \dfrac{1}{f(x)}$ and whether it is a maximum or a minimum, and state the domain and range of $y = \dfrac{1}{f(x)}$.

(a) $f(x) = (x-2)^2 + 4$

(b) $f(x) = -(x+1)^2 + 2$

(c) $f(x) = 2(x-3)^2 - 8$

(d) $f(x) = -\frac{1}{2}(x+4)^2 - 3$

(e) $f(x) = -3(x-1)^2$

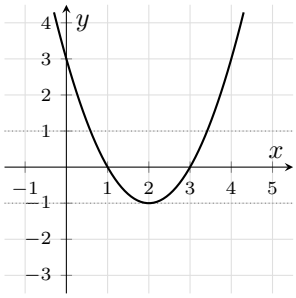
4. Sketching $y = \dfrac{1}{f(x)}$ for a Quadratic

Steps. (1) Graph $y = f(x)$ and draw the lines $y = 1$ and $y = -1$. (2) Draw a dashed vertical asymptote at each zero; the horizontal asymptote is $y = 0$. (3) Mark the invariant points (where f meets $y = \pm 1$), the reciprocal y-intercept $\left(0, \frac{1}{c}\right)$ and the turning point $\left(p, \frac{1}{q}\right)$. (4) Draw each branch on the same side of the x-axis as f: it heads to $\pm\infty$ beside an asymptote, and toward $y = 0$ where $f(x)$ is large.

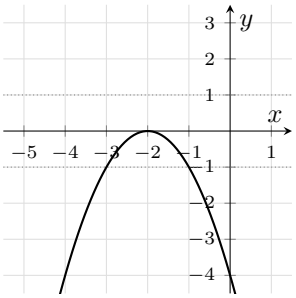
Example 4

The graph of $y = f(x)$ is shown. Sketch the graph of $y = \dfrac{1}{f(x)}$ on the same grid. State the asymptotes and the turning point (if any).

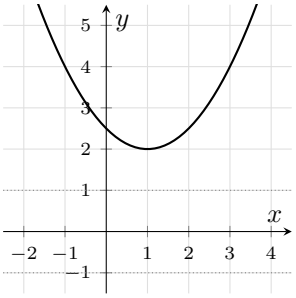
(a) $f(x) = x^2 - 4x + 3$



(b) $f(x) = -(x+2)^2$



(c) $f(x) = \frac{1}{2}(x-1)^2 + 2$

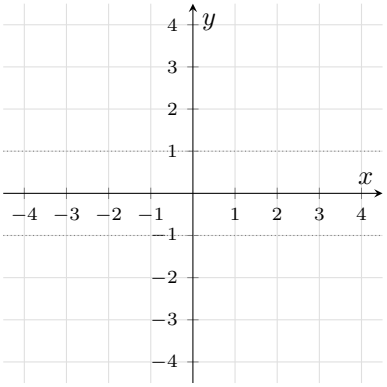


Example 5

Let $f(x) = 2 - \frac{1}{2}x^2$.

(a) Complete the table for $y = \dfrac{1}{f(x)}$. Give exact values.

Vertical asymptotes	
Horizontal asymptote	
y-intercept	
Turning point	
Invariant points	
Domain	
Range	



(b) Sketch $y = f(x)$ and $y = \dfrac{1}{f(x)}$ on the grid. Check with a calculator.

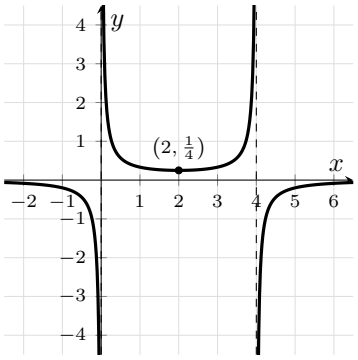
(c) On which intervals is $y = \dfrac{1}{f(x)}$ increasing?

5. Working Backwards from $y = \dfrac{1}{f(x)}$

When the reciprocal graph is given and f is quadratic, reverse the steps: each vertical asymptote $x = a$ gives a zero a of f, a turning point (p, k) gives the vertex $\left(p, \frac{1}{k}\right)$, and a y-intercept $\frac{1}{c}$ gives $f(0) = c$. Write $f(x) = a(x - p)^2 + q$ and use a zero or another point to find a.

Example 6

(a) The graph of $y = g(x)$, where $g(x) = \dfrac{1}{f(x)}$ and f is a quadratic function, is shown. It has vertical asymptotes $x = 0$ and $x = 4$ and a minimum point $\left(2, \frac{1}{4}\right)$.
(i) Sketch the graph of $y = f(x)$ on the same grid.
(ii) Write $f(x)$ in the form $f(x) = a(x - p)^2 + q$, then in general form.



(b) The graph of $y = \dfrac{1}{f(x)}$, where f is quadratic, has a maximum point $\left(-2, \frac{1}{3}\right)$ and a y-intercept of $\frac{1}{5}$. Determine $f(x)$ in vertex form and in general form. Explain why the graph of $y = \dfrac{1}{f(x)}$ has no vertical asymptotes.

6. Points, Matching and Reasoning

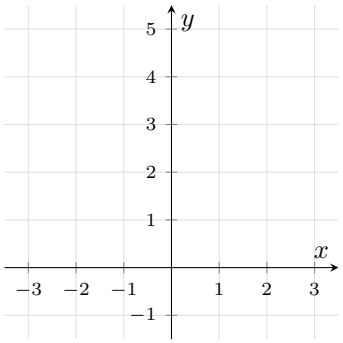
A point (x, y) is on $y = f(x)$ exactly when $\left(x, \frac{1}{y}\right)$ is on $y = \dfrac{1}{f(x)}$, so the conversion works both ways. To find where the reciprocal equals k, solve $f(x) = \dfrac{1}{k}$.

Example 7

(a) The point $(3, 8)$ is on $y = f(x)$. State the matching point on $y = \dfrac{1}{f(x)}$.

(b) The point $(-2, -0.25)$ is on $y = \dfrac{1}{f(x)}$. State the matching point on $y = f(x)$.

(c) Let $f(x) = x^2 + x$. The point $\left(a, \frac{1}{6}\right)$ is on $y = \dfrac{1}{f(x)}$. Find the possible values of a.

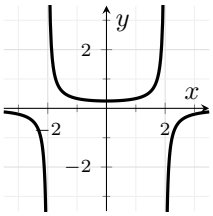


(d) Sketch $y = x^2$ and $y = \dfrac{1}{x^2}$ on the grid. State the asymptotes, the invariant points and the range of $y = \dfrac{1}{x^2}$. Explain why its graph is never below the x-axis.

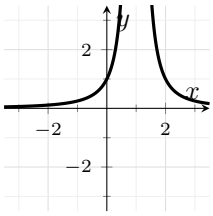
Example 8

Match each function with the graph of its reciprocal $y = \dfrac{1}{f(x)}$. Give one reason for each match.

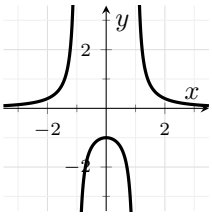
A



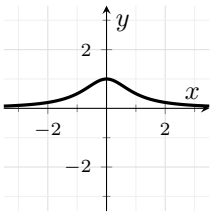
B



C



D



(i) $f(x) = x^2 - 1$ _____

(ii) $f(x) = (x-1)^2$ _____

(iii) $f(x) = x^2 + 1$ _____

(iv) $f(x) = 4 - x^2$ _____

Lesson 8: Unit 7 Review

Absolute value, graphs of $y = |f(x)|$, absolute value equations and reciprocal functions on one page, mixed examples that combine them, and a review assignment written like the unit test.

1. Unit Summary

Topic	Key facts	Example
Absolute value of a number	$\lvert a\rvert$ is the distance from a to 0, so $\lvert a\rvert \geq 0$. $\lvert a\rvert = a$ if $a \geq 0$ and $\lvert a\rvert = -a$ if $a < 0$. The bars are grouping symbols: simplify inside first. The distance between a and b is $\lvert a - b\rvert$.	$\lvert 3-8\rvert - \lvert -2\rvert = 5 - 2 = 3$ Distance from -4 to 7: $\lvert -4 - 7\rvert = 11$
Graph of $y = \lvert f(x)\rvert$	Keep the part of $y = f(x)$ on or above the x-axis; reflect the part below it in the x-axis. The x-intercepts are invariant points. Same domain; the range has no negative values. A line gives a V with its vertex at the zero; for a parabola, the part between or outside the zeros is reflected.	$y = \lvert x^2 - 4\rvert$: the part for $-2 < x < 2$ is reflected; the vertex $(0, -4)$ becomes a local maximum $(0, 4)$; range $\{y \mid y \geq 0, y \in \mathbb{R}\}$
Piecewise form	$\lvert A\rvert = A$ where $A \geq 0$ and $\lvert A\rvert = -A$ where $A < 0$. Solve $A \geq 0$ to find where each piece applies.	$\lvert 2x-6\rvert = \begin{cases} 2x-6, & \text{if } x \geq 3 \\ -2x+6, & \text{if } x < 3 \end{cases}$
Absolute value equations	Isolate the absolute value. $\lvert A\rvert =$ a negative number has no solution. Otherwise solve $A = B$ and $A = -B$ (or use the cases), then verify every answer in the original equation and reject **extraneous** ones. Graphically: the x-coordinates of the intersection points.	$\lvert x-1\rvert = 2x+4$: $x - 1 = 2x + 4$ gives $x = -5$ ($\lvert -6\rvert = 6 \neq -6$: reject); $x - 1 = -(2x+4)$ gives $x = -1$ ($2 = 2$ ✓).

Reciprocal functions $y = \dfrac{1}{f(x)}$. A point (x, y) becomes $\left(x, \frac{1}{y}\right)$. Zeros of $f \to$ vertical asymptotes; the horizontal asymptote is $y = 0$. Invariant points where $f(x) = \pm 1$. Same sign as f. Where f increases, the reciprocal decreases (and the other way). y-intercept $c \to \frac{1}{c}$. Domain: all real numbers except the zeros of f; $y = 0$ is never in the range.

Linear f: one vertical asymptote, two branches in opposite regions; range $\{y \mid y \neq 0, y \in \mathbb{R}\}$.

Quadratic f with vertex (p, q): two, one or no vertical asymptotes (two, one or no zeros). If $q \neq 0$, the turning point is $\left(p, \frac{1}{q}\right)$: a maximum point if $a > 0$, a minimum point if $a < 0$. Range: reciprocals of the range of f, e.g. $f(x) \geq -4$ gives $y \leq -\frac{1}{4}$ or $y > 0$.

Common errors. $|a + b| \neq |a| + |b|$ in general. $-|5| = -5$, not 5. Not isolating the absolute value before splitting into cases. Forgetting to check for extraneous solutions when the other side contains x. Reflecting a whole parabola instead of only the part below the axis. Calling the reciprocal of a minimum a minimum. Drawing the reciprocal graph crossing the x-axis or an asymptote.

2. Absolute Value and Its Graphs

Example 1

(a) Evaluate $|-8| - |3 - 10| + |\sqrt[3]{-27}|$.

(b) Order from least to greatest: $|-2.5|$, $-|-4|$, $|\sqrt{2} - 3|$, $-|1 - 2|$

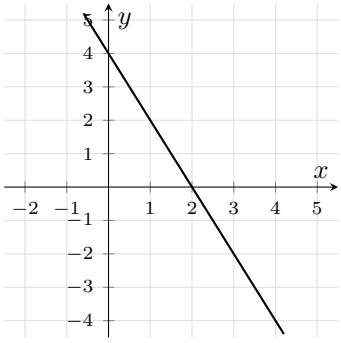
(c) Find the distance between -6.5 and 2.3 on a number line.

(d) Write $y = |5 - 2x|$ in piecewise form.

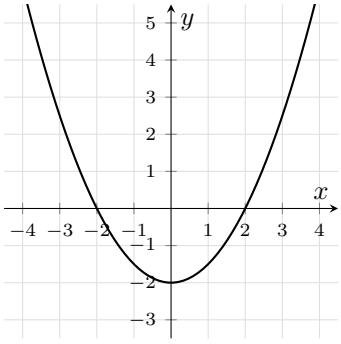
Example 2

The graph of $y = f(x)$ is shown. Sketch $y = |f(x)|$ on the same grid. State the invariant points and the range of $y = |f(x)|$, and write $y = |f(x)|$ in piecewise form.

(a) $f(x) = 4 - 2x$



(b) $f(x) = \frac{1}{2}x^2 - 2$



3. Absolute Value Equations

Example 3

Solve algebraically and verify.

(a) $3|x+2|-4=11$

(b) $|3x-1|+6=2$

(c) $|x+4|=3x-2$

(d) A greenhouse is kept within 3.5°C of 24°C. Write and solve an absolute value equation for the lowest and highest allowed temperatures.

Example 4

Solve algebraically. Verify where needed.

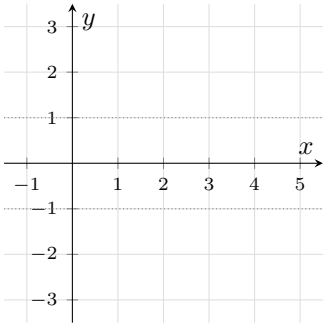
(a) $|x^2-3x-4|=6$

(b) $|x^2-5|=4x$

4. Reciprocal Functions

Example 5

Let $f(x) = 2x - 4$.
(a) Write the equation of the reciprocal function and state its asymptotes, its invariant points and its y-intercept.

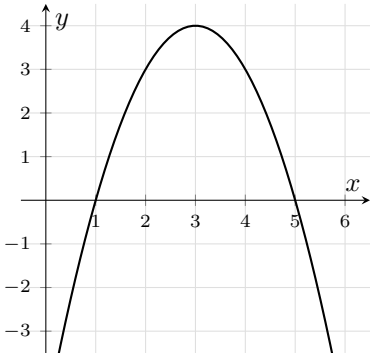


(b) Sketch $y = f(x)$ and its reciprocal on the grid, and state the domain and range of the reciprocal.

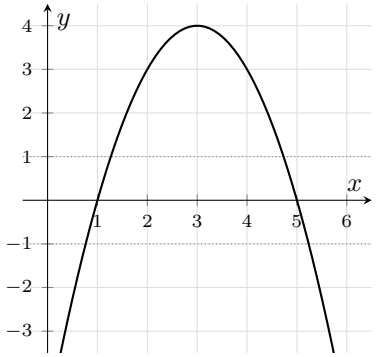
Example 6

Let $f(x) = -x^2 + 6x - 5$. Its graph is shown on both grids.

(a) $y = |f(x)|$



(b) $y = \frac{1}{f(x)}$



(a) Sketch $y = |f(x)|$. State its invariant points and its range.

(b) Sketch $y = \dfrac{1}{f(x)}$. State the asymptotes, the turning point, the exact invariant points, the domain and the range.

Example 7

(a) The point $(-4, -3)$ is on $y = f(x)$. State the matching points on $y = |f(x)|$ and on $y = \dfrac{1}{f(x)}$.

(b) Which of the points $(2, 1)$, $(5, -1)$, $(0, 0)$, $(3, 4)$ on $y = f(x)$ must also be on both $y = |f(x)|$ and $y = \dfrac{1}{f(x)}$? Explain.

(c) A function f is positive and decreasing on the interval $0 < x < 5$. Describe $y = |f(x)|$ and $y = \dfrac{1}{f(x)}$ on that interval.

(d) The graph of $y = \dfrac{1}{f(x)}$, where f is quadratic, has a minimum point $(2, -0.2)$ and no vertical asymptotes. It passes through $(0, -\frac{1}{9})$. Determine $f(x)$.